

Results

The data of 200 participants were used for the analysis. Boxplots of the post-treatment and follow-up timepoints revealed no obvious outlier. Data from all participants was included in the sample to preserve data integrity. The descriptive statistics of the sample can be found in Table 1.

Table 1
Per Condition the Number of Participants, Male to Female to Non-binary Ratio, Mean Age and Standard Deviations (in Parentheses)

	EMDR	In-vivo Exposure Therapy
Number of Participants	100	100
Female/Male/Non-binary	72/24/4	66/30/4
Mean Age	40.79 (14.77)	40.61 (14.79)

A 2 (condition: EMDR vs. In-vivo Exposure Therapy) by 3 (time of measurement: pre- vs. post-assessment vs. three-month follow-up) Factorial Mixed Analysis of Variance (ANOVA) was performed for the mean scores on PCL-5. The assumption of normality was violated based on the Shapiro-Wilk test for the pre-test measurement for the EMDR condition ($W=.95, p=.003$), and for the pre-test measurement for the In-vivo Exposure Therapy condition ($W=.97, p=.023$), the QQ plots also confirmed those deviations from normality. As the sample size was bigger than 30 ($N=200$), the Central Limit Theorem can be applied (Field, 2025) and the analysis was conducted as planned. All other relevant assumptions (e.g. homogeneity of variances, sphericity) were met.

There were a significant main effects of time of measurement, $F(2, 395.56) = 279.14, p < .001$, and condition, $F(1, 198) = 65.08, p < .001$. Importantly a significant interaction was observed between time of measurement and condition, $F(2, 395.56) = 112.20, p < .001$ indicating that the change in PCL-5 scores over time differed per condition. Mean scores per condition and time of measurement can be found in Table 2.

Table 2
Per condition the Mean PCL-5 scores and (Standard Deviations), at Pre-, Post- and 3-Month Follow-up.

	EMDR	In-Vivo Exposure Therapy
Pre-assessment	89.35 (16.55)	87.82 (16.22)
Post-assessment	50.43 (19.27)	67.07 (19.09)
3-month follow-up	49.470 (20.73)	86.63 (19.09)

Holm corrected Post-hoc tests were carried out to examine differences between conditions at and between different time points. Results showed no difference in mean PCL-5 scores between conditions at pre-measurement ($p = 1.000$). Between the pre- and the post-measurement, mean PCL-5 scores decreased significantly in both the EMDR ($p < .001$), and In-vivo condition ($p < .001$). The difference between conditions at post-measurement was significant ($p < .001$), with the EMDR condition showing the biggest reduction in PCL-5 scores compared to pre-measurement (see Figure 1). Between post-assessment and the 3-months follow-up, the scores of the In-vivo condition surprisingly increased significantly ($p < .001$) to the same level as the pre-measurement scores ($p = 1.000$). In contrast, the PCL-5 scores of the EMDR condition did not significantly change post to follow-up ($p = 1.000$). This resulted in a significant difference between the two conditions at follow ($p < .001$), with the EMDR condition showing a reduction in PCL-5 scores compared to pre-measurement which was not visible in the In-vivo group (see Figure 1).

Figure 1

Average score on PCL-5 for Condition (EMDR, in-vivo Exposure Therapy) and Time of Assessment (Pre-, Post-, 3-Month Follow-up).

